

Reviewer Pack — Symplectic Capacity Inverse Proportional to Extended Frege P...

Entry 63a119c032af · INCONCLUSIVE

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Symplectic Capacity Inverse Proportional to Extended Frege Proof Length for 3-SAT

Entry ID: 63a119c032af **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-13 11:47:26 UTC

1. Conjecture

Field A (mathematical branch): Symplectic Geometry **Field B** (complexity object): Extended Frege Proof Length for 3-SAT

Statement:

For a 3-SAT instance Φ with n variables, the symplectic capacity $C(\Phi)$ of the convex hull of its clause vectors in $\mathbb{R}^n$ is inversely proportional to the length of the shortest Extended Frege proof of Φ . Specifically, $C(\Phi) = \Theta(1 / L(\Phi))$, where $L(\Phi)$ is the proof length.

Rationale (proposer's reasoning):

Symplectic capacities capture the 'size' of a convex body in a way that relates to the complexity of symplectic embeddings. By mapping clauses to points in a symplectic space, the capacity could reflect the minimal resources needed to 'embed' the proof, inversely correlating with proof length.

Taxonomy category: PROOF_COMPLEXITY_EXT_FREGE (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): e12779b3eb58490e

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.95	SAFE	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
ALGEBRIZATION	SAFE	0.95	SAFE	UNCERTAIN
KARP_LIPTON	SAFE	0.95	UNCERTAIN	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_3sat_instance(n):
        clauses = []
        for _ in range(2 * n):
            clause = [random.choice([-1, 1]) * (i + 1) for i in range(n)]
            clauses.append(clause)
        return clauses

    def symplectic_capacity(convex_hull):
        # Placeholder for actual computation
        return random.random()

    def extended_frege_proof_length(sat_instance):
        # Placeholder for actual computation
        return len(sat_instance)

    n = 10
    sat_instance = generate_3sat_instance(n)
    convex_hull = [sum(clause) for clause in zip(*sat_instance)]
    capacity = symplectic_capacity(convex_hull)
    proof_length = extended_frege_proof_length(sat_instance)

    return {
        "metric_name": "symplectic_capacity",
        "metric_value": capacity,
        "instances_tested": 1,
        "conjecture_holds": False if capacity == 0 else proof_length > 0 and abs(capacity * proof_length) > 0,
        "counterexample": "mapping_undefined" if capacity == 0 else ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i + 1 for i in range(5, 8)]
```

```

results = []
for seed in seeds:
    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

mean_capacity = sum(r["metric_value"] for r in results) / len(results)
std_capacity = math.sqrt(sum((r["metric_value"] - mean_capacity)**2 for r in results) / len(results))
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_capacity} std={std_capacity} support_fraction={support_fraction}")
elif any(not r["conjecture_holds"] for r in results):
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
    print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")
else:
    print("RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

False, 'counterexample': ''
TRIAL: {'metric_name': 'symplectic_capacity', 'metric_value': 0.44894481047459256, 'instances_tested': 1000}
TRIAL: {'metric_name': 'symplectic_capacity', 'metric_value': 0.9291776936360223, 'instances_tested': 1000}
TRIAL: {'metric_name': 'symplectic_capacity', 'metric_value': 0.8869058471707777, 'instances_tested': 1000}
TRIAL: {'metric_name': 'symplectic_capacity', 'metric_value': 0.14326180295131352, 'instances_tested': 1000}
TRIAL: {'metric_name': 'symplectic_capacity', 'metric_value': 0.6533314106936352, 'instances_tested': 1000}
TRIAL: {'metric_name': 'symplectic_capacity', 'metric_value': 0.5840760431807874, 'instances_tested': 1000}
TRIAL: {'metric_name': 'symplectic_capacity', 'metric_value': 0.772941505723595, 'instances_tested': 1000}
TRIAL: {'metric_name': 'symplectic_capacity', 'metric_value': 0.7209762349959354, 'instances_tested': 1000}
TRIAL: {'metric_name': 'symplectic_capacity', 'metric_value': 0.36527838839309146, 'instances_tested': 1000}

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5ab59ac8.py", line 67, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                        ~~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5ab59ac8.py", line 67, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                        ~~~~~~
KeyError: 'seed'

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test crashed before producing valid data; conjecture_holds=False for multiple trials but no counterexample provided | next: Fix the test code to handle seed tracking and re-run with proper error handling

11. Audit log (LLM calls)

Total LLM calls: 8

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	121034
2	propose	ollama_remote	qwen3:8b	0	0	103600
3	preregistration	ollama_remote	qwen3:8b	0	0	25058
4	novelty	ollama_remote	qwen3:8b	0	0	20608
5	novelty	ollama_remote	qwen3:8b	0	0	14814
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	49230
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6698
8	judge	ollama_remote	qwen3:8b	0	0	18291

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 359334 ms total latency. Provider mix: {'ollama_remote': 8}

(full prompt+response transcripts available in `research/audit/63a119c032af.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/63a119c032af.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/63a119c032af.tar.gz` (if generated)